William Troske

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Education

Ph.D. in Agricultural and Resource Economics, University of California, Davis (Expected) 2027

Committee: James N. Sanchirico, Jamie Hansen-Lewis, Erich Muehlegger

M.S. in Economics, University of Wisconsin, Madison 2022

B.S.E. in Chemical Engineering, University of Michigan 2017

Research Fields

Environmental Economics, Industrial Organization, Agricultural Economics, Trade

Research in Progress

The Behavioral Impact of Sulfur Fuel Content Regulations in Maritime Shipping

Abstract In 2012, the US and Canada implemented a technological regulation, called the ECA (Emission Control Area), in their exclusive economic zones to reduce emissions from ocean-going ships. The ECA required ships to use low-sulfur fuel, increasing a ship's fuel costs. In transportation, however, an increase in fuel cost will lead to changes in shipping supply characteristics such as operating speed, ship size, and frequency. This paper aims to quantify the extent of adjustment in maritime shipping supply characteristics in response to the ECA regulation. I first present a stylized model to highlight how the ECA affects shipping supply characteristics by changing the cost of fuel. I then test these predictions using port entrance and exit data from the US Army Corps of Engineers. I find that for the aggregate cost increase, treated routes saw ships that were, on average, about 15 percent larger in gross tons and had 25 more trips per quarter compared with similar, untreated routes, and ships on treated routes moved 16 percent slower than predicted in the imputed counterfactual. The size of the cost change matters, however, for the response to ship size.

Behavioral Responses to Information on PFAS in Drinking Water
with Laura Alcocer Quiñones and Jeff Hadachek

Abstract This paper estimates the effect of information provision on the awareness and avoidance of PFAS exposure. Past research quantified how individuals avoid exposure to contaminants, such as *E. coli* or lead, when notified by measuring additional bottled water purchases. The case of PFAS is unique; the full health effects of PFAS are relatively uncertain, and PFAS stays in drinking water longer than other common contaminants. We leverage variation in violations across nine states with PFAS regulations between 2019 and 2023 to quantify these impacts. We pair detailed violation data at the weekly level with retail scanner data to study how households avoid contamination by buying bottled water or water filters after receiving notifications about PFAS in their drinking water. Because notifications occurred at different times, we use the imputation difference-in-differences estimator of Borusyak et al. (2024). Our results find bottled water expenditure increases by 7% following the first PFAS violation, with no significant increase in water filter

expenditure. This increase in bottled water expenditure represents an aggregate cost of approximately $4 million for the first PFAS violation.

Early Stage Work

Refrigeration and the Spread of California Agriculture

Pipeline Shutdowns and Jones Act Waivers: Estimating the Cost of the Jones Act

The Effect of Low Water Levels on the Mississippi River and Agricultural Transportation

Academic Publications

"Carious Lesions: Nanoparticle-Based Targeting and Detection of Microcavities" (2017) *Advanced Healthcare Materials* 6(1), 2192-2640. DOI: 10.1002/adhm.201600883 (with Nathan A. Jones, Sywe-Ren Chang, Brian H. Clarkson, Joerg Lahann)

Scholarships, Honors, and Awards

Scott McNiven Community Service Award	2026
Institute for Humane Studies Fellowship	2026
Becker Friedman Institute Price Theory Summer Camp	2025
UC Davis Jastro-Shields Graduate Research Award	2024
UC Davis Summer Research Fellow	2024
UW Madison, Honorable Mention Scholarship (Outstanding Research Paper in Applied Econometrics)	2022
Eagle Scout, Boy Scouts of America	2013

Teaching Experience

Associate Instructor, University of California, Davis

Principles of Finance	Summer 2026

Teaching Assistant, University of California, Davis

Intermediate Microeconomics	Spring 2025, Fall 2024, Fall 2022
Econometric Theory & Applications	Summer 2024
Environmental Economics	Fall 2026, Fall 2025, Winter 2025, Spring 2024
Operations Research & Management Science	Summer 2023
Principles of Finance	Spring 2026
Economic & Business Analytics	Winter 2026

Research Experience

Research Assistant, University of California, Davis

Tim Beatty & Joakim Weill: Estimated the labor response to wildfire smoke	Fall 2023 - Winter 2024

Research Assistant, University of Wisconsin, Madison

Sarah Johnston: Created an algorithm in R to match car information to VINs — Summer 2021

Research Assistant, University of Michigan, Ann Arbor

Nathan Jones: Developed a starch nanoparticle to identify carious lesions — Fall 2015 - Spring 2017

Professional Activities

UCD Natural Resource Economics and Policy Lab

Graduate Student Coordinator — Fall 2025 - Fall 2026

UCD ARE Graduate Student Association

Representative — Fall 2022 - Fall 2024

Co-President — Fall 2024 - Fall 2025

Policy Researcher - Clean Wisconsin — March 2022 - Aug 2022

Summarized current academic knowledge around natural climate solutions (e.g., agroforestry or cover crops) for carbon sequestration in agriculture, and made estimates of their effectiveness in Wisconsin

Quoting and Process Engineer - Advanced Plating Technologies — Jan 2018 - July 2020

Served as an engineering resource and initial contact for customers in need of surface finishes for their parts.

Presentations and Workshops

Agricultural & Applied Economics Association Annual Meeting — July 2026

Western Economic Association International Annual Conference — July 2026

Giannini Foundation of Agricultural and Resource Economics Student Conference — May 2025 & May 2026

8th Annual Social Cost of Water Pollution Conference — Sept 2025

Other Writing and Media

Advanced Plating Technologies - Technical Blog

Solderable Gold Plating of Electrical Contacts (Aug 6th, 2020)

Nitric vs Citric Acid Passivation (July 11th, 2019)

Other

Software

Stata, R, Python, Julia, Git, ImageJ

References

Dr. James N. Sanchirico (Chair)
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Environmental Science and Policy
University of California, Davis
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Dr. Erich Muehlegger
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Economics
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Dr. Jamie Hansen-Lewis
Assistant Professor
Agricultural & Resource Economics
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